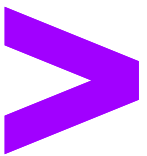


Switching To Windows & Frames



Module Objective

At the end of this course, you will be able to:

- Understand about native dialogs
- Demonstrate how to handle native dialogs
- Demonstrate how to handle multiple windows
- Understand about Iframes
- Demonstrate how to handle Iframes

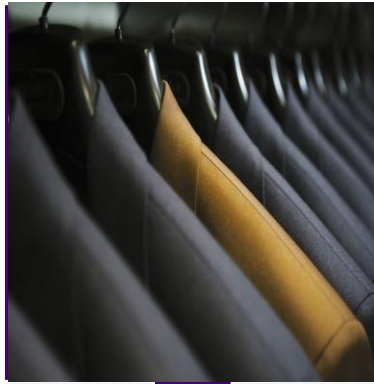
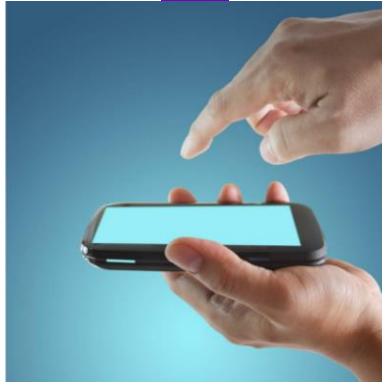
Ground Rules

For a successful class, please:



Arrive on time

Keep your cell phones off



Wear business formals

Assist and show respect to your colleagues irrespective of skill and knowledge level



Do not use the web for personal reasons during training sessions

Adhere to attendance policy as directed by training coordinator



Switching to Windows & Frames

- 01** Introduction to native dialogs
- 02** Handle Native dialogs
- 03** Handle Multiple Windows
- 04** Handle frames

Introduction to Native Dialogs

Introduction to Native Dialogs (1)

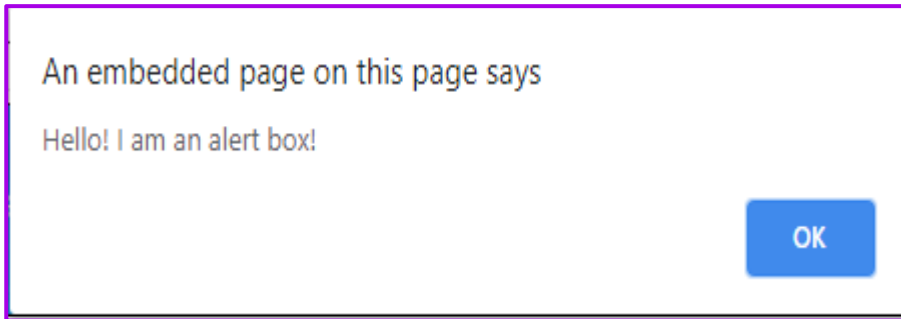
Native Dialogs

- A native dialog is an HTML container to show content on the web page
- It is in the foreground and overlaps the rest of the page
- The main intention of a developer to design an HTML dialog is to get the attention of the user and forces the user to take certain action
- It is designed using JavaScript and blocks the interaction with the rest of the web page
- **Examples of Native dialogs:**
 - Alert Dialog Box
 - Confirmation Dialog Box
 - Prompt Dialog Box

Introduction to Native Dialogs (2)

Types of Native Dialogs

Alert Dialog



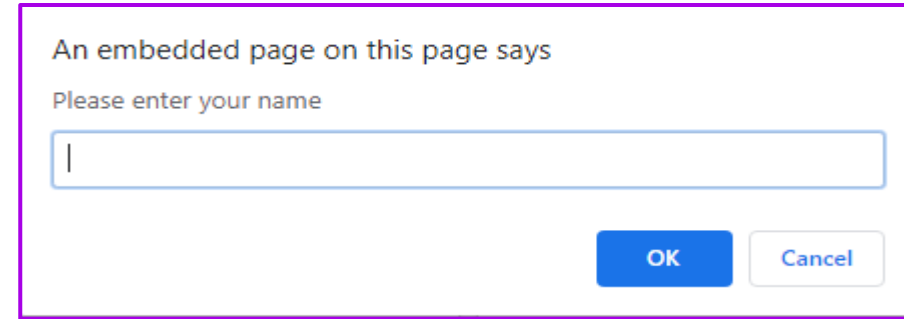
An embedded page on this page says

Hello! I am an alert box!

OK

This is a screenshot of an alert dialog box. It has a white background and a thin purple border. The text inside says "An embedded page on this page says" followed by "Hello! I am an alert box!". At the bottom right, there is a single blue button with the text "OK".

Prompt Dialog



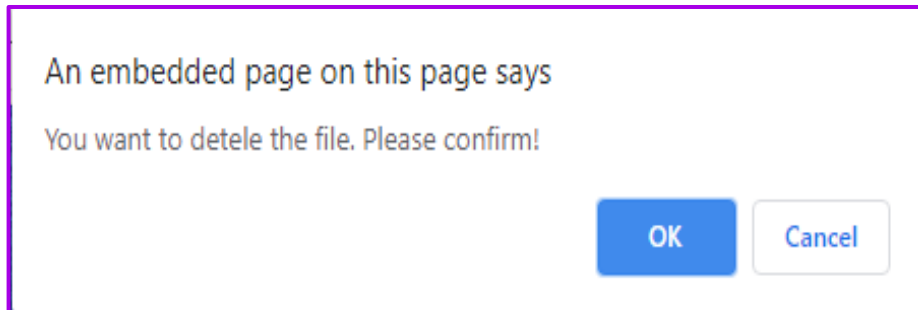
An embedded page on this page says

Please enter your name

OK Cancel

This is a screenshot of a prompt dialog box. It has a white background and a thin purple border. The text inside says "An embedded page on this page says" followed by "Please enter your name". Below the text is a text input field with a light blue border. At the bottom right, there are two buttons: a blue "OK" button and a white "Cancel" button with a light blue border.

Confirm Dialog



An embedded page on this page says

You want to detele the file. Please confirm!

OK Cancel

This is a screenshot of a confirm dialog box. It has a white background and a thin purple border. The text inside says "An embedded page on this page says" followed by "You want to detele the file. Please confirm!". At the bottom right, there are two buttons: a blue "OK" button and a white "Cancel" button with a light blue border.

Handle Native dialogs

Handle Native Dialogs (1)

Alert Class in WebDriver

- Selenium WebDriver must handle the native dialogs like alert, confirmation, and prompt, generated by the web application to ensure a smooth flow in the transaction
- Selenium WebDriver provides an **Alert class** for working with dialog box generated by JavaScript
- To handle the dialog box:
 - Check the presence of an alert, confirmation, or prompt dialog on the web page
 - Interact with the dialog box and perform the next action
- **ExpectedConditions.alertIsPresent()** can be used to check the presence of a dialog box in the automation flow of events

Handle Native Dialogs (2)

Handle Alert Dialog

- Check the presence of an alert
- Use **switchTo()** method to interact with the Alert Dialog
- Use **getText()** method to fetch the text on Alert Dialog
- Use **accept()** method to click on the OK button



```
if(ExpectedConditions.alertIsPresent()!=null) {  
    Alert alert = driver.switchTo().alert();  
    String text = alert.getText();  
    System.out.println("Alert dialog text is: " + text);  
    alert.accept();  
}
```

Handle Native Dialogs (3)

Handle Confirmation Dialog

- Check the presence of an alert
- Use **switchTo()** method to interact with the Confirmation Dialog
- Use **accept()** method to click on the OK button



```
if(ExpectedConditions.alertIsPresent()!=null) {  
    Alert alert = driver.switchTo().alert();  
    alert.accept();  
}
```

- Check the presence of an alert
- Use **switchTo()** method to interact with the Confirmation Dialog
- Use **dismiss()** method to click on the Cancel button



```
if(ExpectedConditions.alertIsPresent()!=null) {  
    Alert alert = driver.switchTo().alert();  
    alert.dismiss();  
}
```

Handle Native Dialogs (4)

Handle Prompt Dialog

- Check the presence of an alert
- Use **switchTo()** method to interact with the Prompt Dialog
- Use **sendKeys()** method to type text in the Prompt
- Use **accept()** method to click on the OK button



```
if(ExpectedConditions.alertIsPresent()!=null) {  
    Alert alert = driver.switchTo().alert();  
    alert.sendKeys("James");  
    alert.accept();  
}
```

- Check the presence of an alert
- Use **switchTo()** method to interact with the Prompt Dialog
- Use **dismiss()** method to click on the Cancel button



```
if(ExpectedConditions.alertIsPresent()!=null) {  
    Alert alert = driver.switchTo().alert();  
    alert.dismiss();  
}
```

Handle Native Dialogs (5)

Demo Solution: Automate a scenario with native dialog (1)

Run ALL

```
public class AlertHandling {
    WebDriver driver;

    @BeforeTest
    public void setUp() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        driver.get("https://lkmdemoaut.accenture.com/AccenSportz/#!/actions/playerandteamstrategy");
    }

    @AfterTest
    public void tearDown() {
        // Close the browser
        driver.quit();
    }
}
```

Handle Native Dialogs (6)

Demo Solution: Automate a scenario with native dialog (2)

```
@Test
Run | Debug
public void alert() {
    Actions act = new Actions(driver);
    act.moveToElement(driver.findElement(By.xpath("//*[@id=\"menu3\"]/li[3]/a/span/b")))
        .moveToElement(driver.findElement(By.id("united"))).click().build().perform();

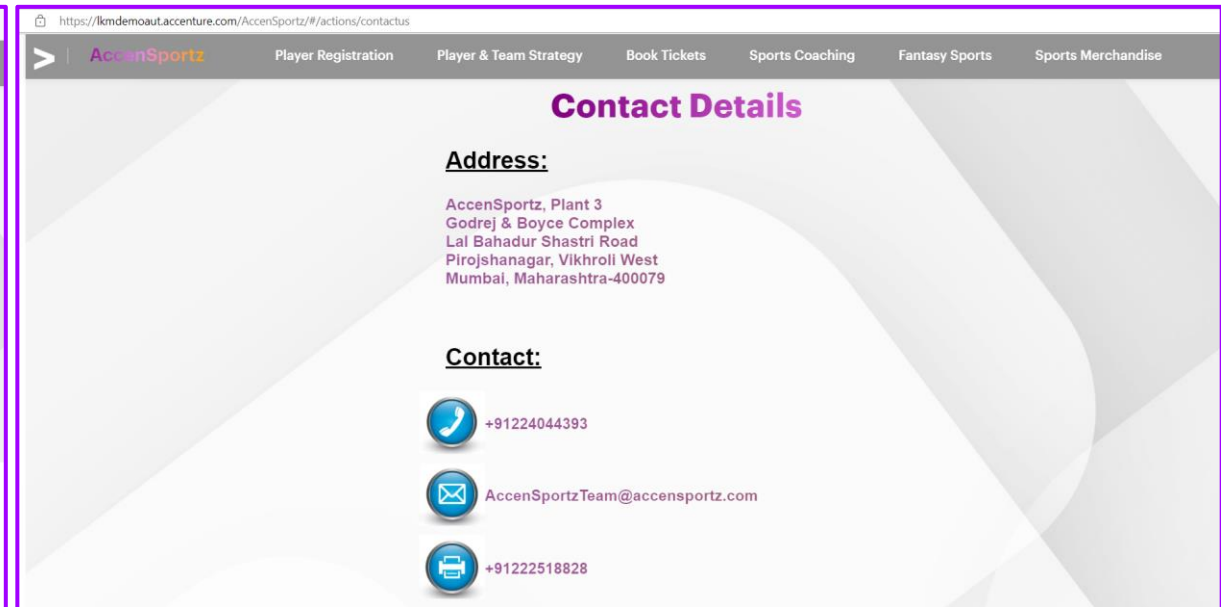
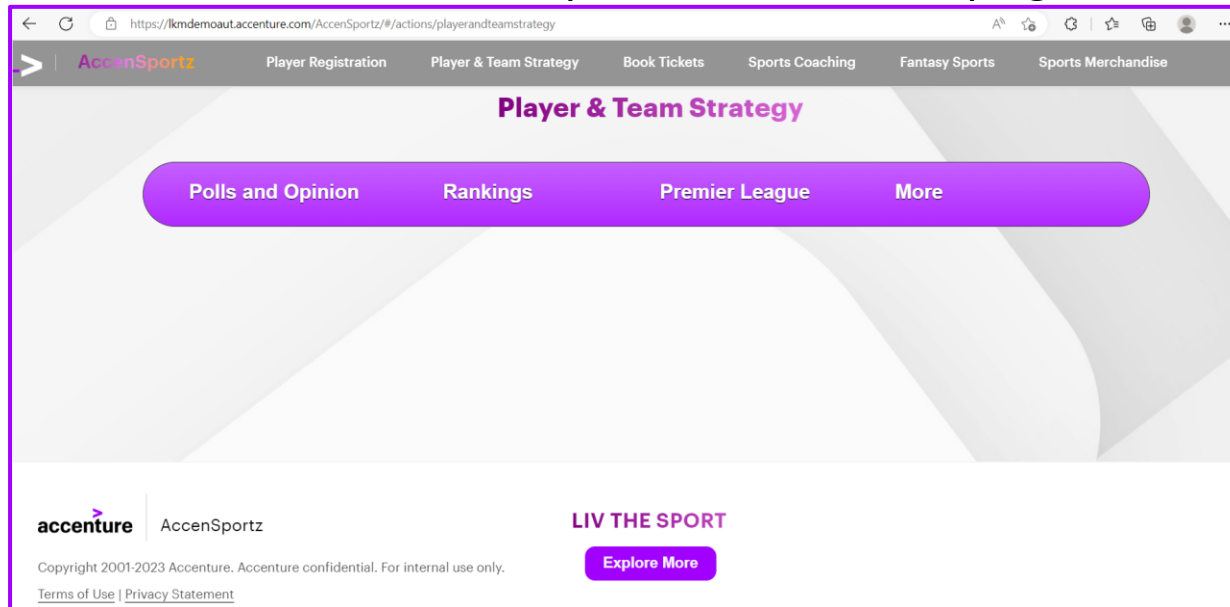
    // handling Alert
    Alert al = driver.switchTo().alert();
    // getting text of an alert
    String alertTxt = al.getText();
    System.out.println("alert text is:" + alertTxt);
    // accepting the alert
    al.accept();
}
```

Handle Multiple Windows

Handle Multiple Windows (1)

Multiple Windows

- In certain applications, when a user takes an action or clicks on a link/button, the application page opens in a new window/tab
- The automation tool must have a capability to handle such a situation and interact with the page elements on the new window/tab
- Example: In AccenSportz application, click on **Player & Team Strategy -> More -> information-> Contact Us**. This opens the contact page in a new tab



Handle Multiple Windows (2)

Interact with Multiple Windows

- Selenium WebDriver creates a unique String Id to each window when the instance of WebDriver is created
- This unique String Id is called “window handle”
- Selenium uses the String Id to switch control between the windows
- It can differentiate windows based on the unique String Id when switching from one window to another

Handle Multiple Windows (3)

Interact with Multiple Windows

1. **getWindowHandles()** method can fetch all the window instances opened by the WebDriver object

```
Set<String> allWindows = driver.getWindowHandles();
```

2. **getWindowHandle()** method can fetch the current window instance opened by the WebDriver object

```
String windowName = driver.getWindowHandle();
```

3. **switchTo().window()** method can switch the WebDriver object to another window by specifying the window handle and for loop allows iterating over every open window

```
for(String handle : driver.getWindowHandles()) {  
    driver.switchTo().window(handle);  
}
```



Handle Multiple Windows (4)

Demo: Automate a scenario with multiple windows

- James wants to automate a scenario which involves web pages in multiple windows
- Scenario

1.

Open AccenSportz URL: <https://lkmdemoaut.accenture.com/AccenSportz/#/> on a web browser

2.

Navigate to Contact us page from player & team strategy

3.

Verify AccenSportz text on the child window of player & team stratgey

4.

Close the browser

Handle Multiple Windows (5)

Demo Solution: Automate a scenario with multiple windows (1)

```
package windowHandling;

import java.time.Duration;
import java.util.Set;

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;
import org.testng.Assert;
import org.testng.annotations.AfterTest;
import org.testng.annotations.BeforeTest;
import org.testng.annotations.Test;
```

```
public class WindowHandling {
    WebDriver driver;
    @BeforeTest
    public void setUp() {
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        driver.get("https://lkmdemoaut.accenture.com/AccenSportz/#/");
    }
    @AfterTest
    public void tearDown() {
        // Close the browser
        // driver.quit();
    }
}
```

Handle Multiple Windows (6)

Demo Solution: Automate a scenario with multiple windows (2)

```
@Test
Run | Debug
public void f() throws InterruptedException {

    driver.findElement(By.xpath("//a[@class='playerandteamstrategy']")).click();
    Actions act=new Actions(driver);
    act.moveToElement(driver.findElement(By.xpath("//b[normalize-space()='More']")))
        .moveToElement(driver.findElement(By.xpath("//span[@name='info']")))
        .moveToElement(driver.findElement(By.xpath("//span[@id='contactus']")))
        .click().build().perform();
    //verify text on alertbox
    String alerttext=driver.switchTo().alert().getText();
    if(alerttext.contains("Please enter your contact details")) {
        driver.switchTo().alert().sendKeys("saranya");
        driver.switchTo().alert().accept();
    }
    Thread.sleep(20000);

    String parentwindow = driver.getWindowHandle();
    Set<String> windows = driver.getWindowHandles();

    Iterator<String> ite = windows.iterator();
    while(ite.hasNext()) {
        String childwindow = ite.next();
        if(!parentwindow.equalsIgnoreCase(childwindow)) {
            driver.switchTo().window(childwindow);
            System.out.println(driver.getWindowHandle());
            //verify AccenSportz text
            String text=driver.findElement(By.xpath("//h4[contains(text(),'AccenSportz, Plant 3')]")).getText();
            Assert.assertTrue(text.contains("AccenSportz"));
        }
    }
}
```

NEW IN SELENIUM 4

New Window and Tab

- Selenium 4 makes it easier for users to handle a browser window and tab
- The use of `newWindow()` method allows the driver to open a new window/tab and automatically switch
- A new window or tab can be created by specifying the `WindowType` in the `newWindow()` method
 - `WindowType` to create a new tab is **TAB**
 - `WindowType` to create a new window is **WINDOW**

```
// Driver creates and switches to the New Tab
driver.switchTo().newWindow(WindowType.TAB);

// Driver creates and switches to the New Window
driver.switchTo().newWindow(WindowType.WINDOW);
```

Handle Frames

Handle Iframes (1)

Iframe

- Iframe is a HTML code, which can be used to embed a web page inside another web page
- It can embedded a HTML page, PDF page, or a web site into another webpage inside a window
- The window can be styled using CSS
- **Example of iframe**
 - `<iframe src="demoIFrame.htm" style="height:250px;width:250px;"></iframe>`

HTML IFrames

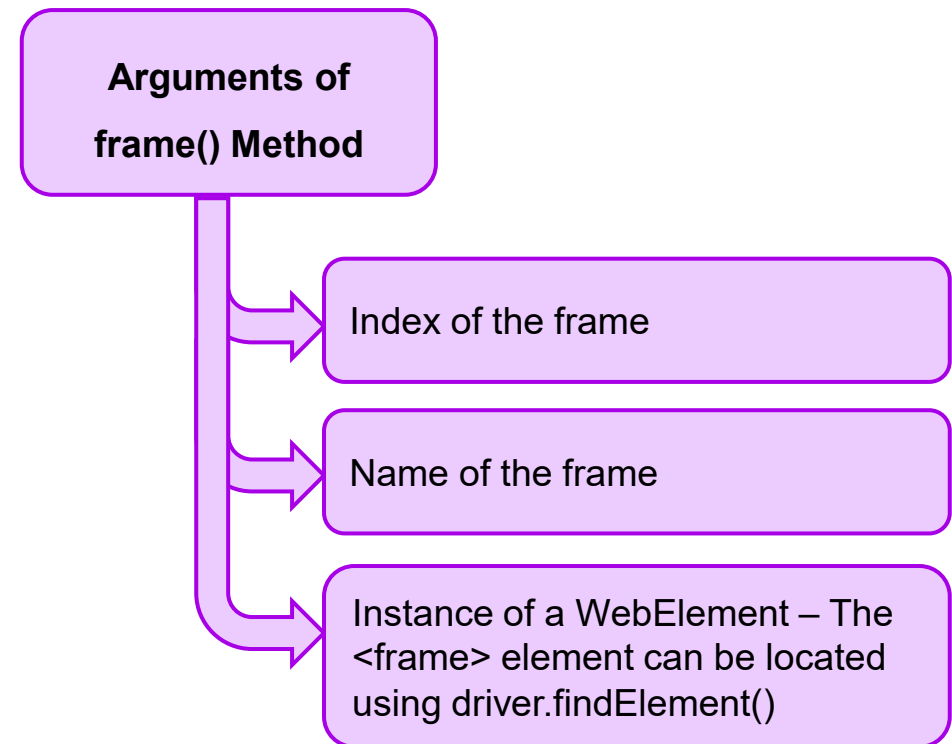
This is the main web page

This is the IFrame

Handle Iframes (2)

Identifying and Handling Iframe

- **List<WebElement> frames = driver.findElements(By.tagName("iframe"));**
- The above line of code gets all the frames on the page created with <iframe> tag
- The Selenium WebDriver API provides a method to switch to a frame on a web page and perform operations using the driver instance
- Method: **driver.switchTo().frame()**
- Once the frame is activated, the driver instance will allow operations on the document loaded in the frame



Handle Iframes (3)

Techniques to interact with Iframes

- **Name** of the frame can be used to switch the driver to a particular frame if the name attribute is defined in the <iframe>

```
driver.switchTo().frame("frameName");
```

- **Index** of the frame can be used to switch the driver to a particular frame. This technique is very useful when Id or Name attribute is not defined in the <iframe>

```
driver.switchTo().frame(0);
```

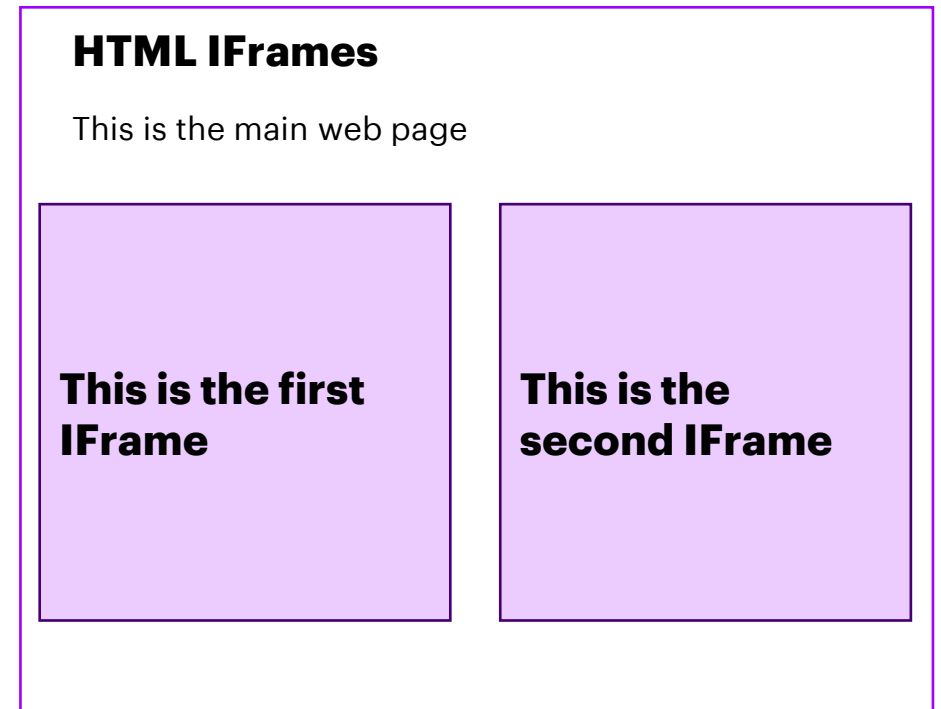
- Handling frames by the index is not reliable when application UI is dynamic. A better technique of handling frames is by identifying frame as a **WebElement**

```
driver.switchTo()  
.frame(driver.findElement(By.id("frameId")));
```

Handle Iframes (4)

Multiple Iframe Elements

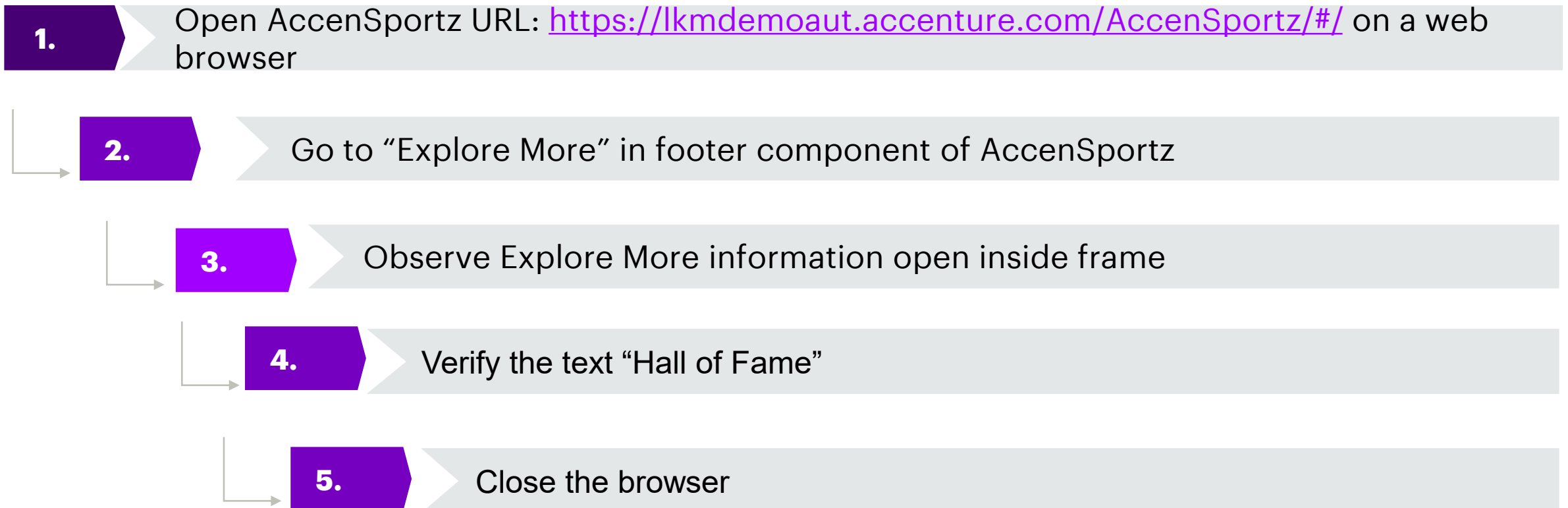
- A web page may be designed by UI developer with multiple iframe elements
- Working with multiple iframes:
 - Navigate to one iframe and perform the desired operation
 - Activate the main document by calling **driver.switchTo().defaultContent()**
 - Navigate to another iframe and perform the desired operation
- **Note:** Driver cannot navigate from one iframe to another iframe directly. Driver must activate the main document and then activate the desired iframe



Handle Iframes (5)

Demo: Automate a scenario with Iframe

- James wants to automate a scenario which involves web page having iframe element
- Scenario



Handle Iframes (6)

Demo: Automate a scenario with Iframe

```
@Test
Run | Debug
public void f() {
    driver.findElement(By.xpath("//button[@id='exploremore']")).click();
    //switch to frame1
    driver.switchTo().frame(driver.findElement(By.id("frame1")));
    //switch inside frame
    driver.switchTo().frame(driver.findElement(By.id("frame2")));
    String frametext=driver.findElement(By.id("text2")).getText();
    Assert.assertTrue(frametext.contains("Hall of Fame"));
}
```

```
@BeforeMethod
public void beforeMethod() {
    // open the Chromebrowser
    driver = new ChromeDriver();
    driver.manage().window().maximize();// maximizing browser
    driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
    driver.get("https://lkmdemoaut.accenture.com/AccenSportz/#/");
}

@AfterMethod
public void afterMethod() throws InterruptedException {
    driver.quit();
}
```

Knowledge Check(1)

Select the correct answers



Which of the following methods will be used to handle an Alert dialog box?

a) alert.accept()

b) alert.cancel()

c) alert.ok()

Knowledge Check(2)

Select the correct answers



Which method is used in Selenium WebDriver to switch to a new window?

1.driver.switchTo.Window()

2.driver.getWindoHandle()

3.driver.navigate.To()

Knowledge Check(3)

Select the correct answers



Which of the following can be a valid way to identify an iframe?

1.By index

2.By name

3.Both 1 & 2

Module Summary

Now, you will be able to:

- Understand about native dialogs
- Demonstrate how to handle native dialogs
- Demonstrate how to handle multiple windows
- Understand about Iframes
- Demonstrate how to handle Iframes



Thank You

